

```
In [2]: import pandas as pd

# Create two Series
s1 = pd.Series([10, 20, 30], index=['A', 'B', 'C'])
s2 = pd.Series([5, 15, 25], index=['B', 'C', 'D'])

# Step 1: Add both Series
result1 = s1 + s2
print("After Addition (NaN will appear):\n", result1)

# Step 2: Observe NaN values
print("\nCheck NaN values:\n", result1.isnull())

# Step 3: Replace NaN with 0 and recompute
result2 = s1.add(s2, fill_value=0)
print("\nAfter replacing NaN with 0:\n", result2)
```

After Addition (NaN will appear):

```
A      NaN
B    25.0
C    45.0
D      NaN
dtype: float64
```

Check NaN values:

```
A      True
B    False
C    False
D      True
dtype: bool
```

After replacing NaN with 0:

```
A    10.0
B    25.0
C    45.0
D    25.0
dtype: float64
```

In []: